

Aditya Gangadhar Shinde

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EDUCATION

Syracuse University, Syracuse, NY || MS in Mechanical and Aerospace Engineering

Dec 2026

Mumbai University, Mumbai, India || Bachelor of Engineering in Mechanical Engineering

May 2024

SKILLS

CAD & Design: SolidWorks (proficient), CATIA, AutoCAD, Fusion 360, NX, Creo, 3D Modeling, Sheet-Metal Design, GD&T (ASME Y14.5)

Mechanical Design: Mechanism & assembly design, component specifications, tolerancing, troubleshooting, Root cause Analysis, DFMA,

Analysis: ANSYS, FEA, ANSYS Fluent, CFD, thermal & structural analysis, ANOVA, Taguchi Methods, Six Sigma

Manufacturing: 3D Printing, CNC Machining, Welding Fundamentals, ceramics/high-temperature processing

Documentation: Technical reports, fabrication drawings, Project management

Software: MS Office (Excel, PowerPoint, Word), statistics, Python, MATLAB, Arduino

Soft Skills: Cross-functional work, team player

WORK EXPERIENCE

Electromechanical Design Engineering Intern, D&D Motor Systems Inc, Syracuse, NY

Jan 2026 – May 2026

- Created 2D drawings and 3D models of AC and DC motor parts in SolidWorks and AutoCAD, supporting design updates and manufacturing for the motor product line.
- Tested DC motors on the motor testbed to check they met performance requirements, while also helping build a new testbed for AC motors that wasn't ready yet.
- Designed and built a sheet-metal box for an electrical enclosure did the SolidWorks design myself, then cut, drilled, and assembled it using shop tools.

Research Assistant, Environmental and Controls Lab, Syracuse University, Syracuse NY

Aug 2025 – Dec 2025

- Engineered conductive PPy/PEO nanofiber mats via electrospinning for gas sensing applications, optimizing polymer composition (5 wt% PPy, 2 wt% PEO and solvent system (DI water/DMF, 1:1) to eliminate beading and jet instability.
- Systematically optimized electrospinning process parameters (14–15 kV voltage, 1 mL/h flow rate, 15 cm tip to collector distance) and environmental controls (<50% RH, 20–25°C) to produce uniform 200–250 nm diameter nanofibers verified by SEM.
- Achieved functional semiconducting nanofibers ($\sim 10^{-3}$ S/m via four-point probe) with demonstrated gas-sensing response up to 5000 ppm CO₂, validating the material for conductive polymer sensor applications.

Soft Service Provider, Good Treat, Mumbai, India

June 2022 – Aug 2022

- Prepared process flow diagrams and plant layouts with detailed engineering drawings for fabricated equipment using AutoCAD and SolidWorks, improving design accuracy by 20% and reducing design revision cycles by 15%.
- Developed cost and time tracking charts in MS Excel for recording technical and commercial data, streamlining reporting and achieving a 25% faster data analysis process.

Design and Manufacturing Intern, Perfect Fabricators and Engineers, Mumbai, India

June 2021 – May 2022

- Cut design cycle time 20% on a thermally demanding high pressure Nitrogen Chamber by leading SolidWorks design from system requirements through fabrication drawings, defining materials, tolerances, and finishes including review of end of life and currently available component options during material selection.
- Achieved 100% process compliance and a 15% reduction in production delays by coordinating material selection and process planning with manufacturing, QA, and design teams, and maintaining system documentation and design standards throughout the build.

PROJECTS

Optimization of Turning Operation for EN24 Steel Alloy using Taguchi Techniques

- Optimized machining parameters (cutting speed, feed rate, depth of cut) for EN24 steel alloy using Taguchi experimental design, directly applicable to precision component manufacturing; applied ANOVA to statistically evaluate parameter influence on surface finish and tool wear, identifying key process drivers for quality improvement.

Tripod Design

- Applied Nelder-Mead and Exterior Penalty algorithms in MATLAB, achieving 100% convergence consistency and improving design feasibility by 30%, to design and optimize a structural assembly meeting all stress and buckling constraints while reducing mass by 35 - 40% (to 6.6 kg), demonstrating concurrent optimization for weight, efficiency, and cost targets.

Mathematical Modeling of Cantilever Beam Under Load.

- Developed a Timoshenko-Ehrenfest beam FEA model in MATLAB (100 elements, 200 DOF) to analyze cantilever beam dynamics under free vibration, performing modal analysis achieving <3% error against experimental accelerometer data; implemented Rayleigh damping and modal superposition for transient response simulation, demonstrating ability to validate simulation results against physical test data..

Go-Kart Suspension System Design & Structural Validation

- Designed a double-wishbone rear suspension system for a 180 kg go-kart, deriving spring/damper specifications (150 lb/in, 1,832 N·s/m) from a target 3.0 Hz ride frequency and 0.65 damping ratio using first-principles vehicle dynamics calculations.
- Modeled a full suspension assembly in SolidWorks and structurally sized the lower wishbone in Aluminum 6061-T6 against a calculated 1,777 N worst-case load (3g bump + 1.5g cornering), achieving a 2.90 safety factor validated by hand-calculation and FEA.
- Performed kinematic analysis of camber gain through wheel travel via a four-bar linkage solve, benchmarked design geometry against commercial go-kart components, and delivered a complete design package from load-case derivation through CAD and structural validation.